PR 2 — Ellipsoidal decomposition sampler: implementation of paper §4 and its validation

Bayesian-Reflex implementation (builds on merged PR 1)

Scope in one sentence. PR 2 implements the paper's exact i.i.d. sampler based on ellipsoidal decomposition (arXiv 2608.00492, §4, eqs. 11–13) as a standalone, tested building block, and validates it against closed-form references and against the PR 1 scalar Gaussian belief.

Not covered. The reflex loop, Thompson sampling (§3.3, §6.2), recursive Gaussian processes (§5), derivative-aware optimisation (§6), scaling beyond $d = 5$, and any function-space or PC-FSVI work or comparison with other inference methods (deferred by the project leader). Nothing merged in PR 1 was modified.

1 Continuity with merged PR 1

Merged in PR 1	Tests	Used here
`ScalarGaussianBelief`: exact conjugate update (paper eq. 9, Gaussian case)	E1 33/33, E2a 9/9	oracle for the posterior the sampler must reproduce; source of the partition's (μ, σ^2)
`GenerativeModel` interface; its docstring names the ellipsoidal sampler as a later stage	27 unit tests	`sample(n,rng)` call shape; returns `(n,d)`
`PredictiveCodingLayer`, `BayesianReflex` (error is diagnostic only)	E3 41/41	untouched
Method rules: criteria fixed in advance, negative controls, 10 seeds, SBC, failure exit status	`experiments/ e1–e3`	followed by E4; SBC reused with the sampler as posterior sampler

Why this is next. The paper's belief updates and Thompson sampling are exact *because* the posterior can be sampled exactly (§4.4, §5.5, §6.2); the merged code only has the conjugate case, where sampling is trivial.

2 The method

Let $\tilde{\pi}$ be an unnormalised density on $\mathbb{R}^d$. For μ, Σ and radii $0 = \rho_0 < \rho_1 < \cdots$, the shells are

$$A_i = \{\theta : \ \rho_{i-1}^2 \le (\theta-\mu)^\top \Sigma^{-1} (\theta-\mu) \le \rho_i^2\}, \qquad \pi = \sum_i \pi(A_i)\, \pi_i, \quad \pi_i = \frac{\pi\, \mathbf{1}_{A_i}}{\pi(A_i)}. \tag{1}$$

Weights. $\pi(A_i) \propto \mathcal{L}(A_i)\, \mathbb{E}_U[\tilde{\pi}]$ with U uniform on A_i and $\mathcal{L}(A_i) = |B| V_d(\rho_i^d - \rho_{i-1}^d)$, $B = \mathrm{chol}(\Sigma)$; estimated once per shell from N_i uniform draws.

Perfect draw from π_i. Metropolis–Hastings on A_i with uniform independence proposals Q_i has acceptance $a = \min\{1, \tilde{\pi}(\theta')/\tilde{\pi}(\theta)\} \geq p_i = \inf \tilde{\pi} / \sup \tilde{\pi}$, so $P_i \geq p_i Q_i$ (eq. 13) and $P_i = p_i Q_i + (1 - p_i) R_i$. In a stationary chain the last regeneration before time 0 happened T steps earlier with $P(T = k) = p(1-p)^k$, $k = 0, 1, 2, \ldots$, and only R_i was used afterwards, so

$$\theta^{(0)} = (\text{draw from } Q_i, \text{ then } T \text{ steps of } R_i) \sim \sum_{k \geq 0} p(1-p)^k \, Q_i R_i^k = \pi_i. \tag{2}$$

Residual step. Since $a \geq p$, "accept iff $U < a$" splits into $\{U < p\}$ (probability p; always accepted; new state $\sim Q_i$) and its complement. Conditioning on $U \geq p$ gives R_i exactly: propose $\theta' \sim Q_i$, draw $U \sim \text{Uniform}(p, 1)$, move iff $U < a$. This equals the source's rejection scheme without needing a Monte Carlo estimate of the rejection probability.

Overall sampler (§4.3). Choose shell i with probability $\propto \hat{\pi}(A_i)$, then draw perfectly from π_i; if the last of the M shells is chosen, double M and repeat the choice with the same uniforms.

3 Implementation

Files added (nothing modified): `src/bayesian_reflex/sampling/` (`ellipsoidal.py`, `bounds.py`, `__init__.py`), `tests/test_ellipsoidal.py`, `experiments/e4_ellipsoidal_sampler.py`, `docs/ellipsoidal_sampler.md`, results.

Paper / source	Code
shells (eq. 11), radii $\rho_i = r + a(i-1)$	`EllipsoidalPartition`, `LinearRadii`
shell volume; uniform sampling on a shell	`log_volume`; `sample_uniform` (exact radial inverse CDF, log space)
weights (eq. 12); $\hat{p}_i = \hat{s}_i / \hat{S}_i - \eta$	`EllipsoidalSampler._ensure_shell`
split chain, perfect draw (eq. 13)	`_perfect_sample_shell`
i.i.d. sampler (§4.3), doubling of M	`sample`, `sample_from_shells`
exact shell bounds for Gaussian targets (not in the paper; separates "algorithm" from "Monte Carlo bounds")	`gaussian_shell_log_bounds`

4 Ambiguities and discrepancies

Sec. 4 is one page and defers details to Bhattacharya, arXiv:2107.05956 (the *source*), which was read to fill gaps. None of the following was resolved silently; items 1 and 2 change results and need a decision.

1. **Support of the coalescence time (discrepancy).** §4.2 says only "Geometric". The source (eq. 24) states $t = 1, 2, \ldots$ and says the form starting at 0 failed. With $T \geq 1$ the output law is $(\pi_i - pQ_i)/(1-p)$, not π_i. Checked on a shell with known answer ($p = 0.458$, 200,000 draws): support $\{0, 1, \ldots\}$ gives KS $p = 0.29$ and $\mathbb{E}|x| = 1.2621$ (exact 1.2618); support $\{1, 2, \ldots\}$ gives KS $p \approx 0$ and 1.2292, equal to the predicted $(1.2618 - 0.458 \cdot 1.3)/(1 - 0.458) = 1.2292$. Default: $\{0, 1, \ldots\}$; `geometric_support=1` reproduces the stated form. The source's code could not be tested against its text.
2. **p_i is estimated, not bounded.** The source uses the min and max of $\tilde{\pi}$ over the Monte Carlo draws minus η; these are optimistic, so the minorization can fail and the residual step is then clamped. Default as in the source; violations are *counted* (0 in all Gaussian runs, 2

in 450,779 steps for Student-$t(5)$); rigorous bounds can be supplied (provided for Gaussian targets). The paper gives no rigorous bound for general targets.

3. **Radii and M.** Only a family with experiment-specific constants is given; required arguments here, doubling rule as in Algorithm 1.
4. **μ and Σ.** The source obtains them from TMCMC runs (not implemented); supplied by the caller. Any choice is valid; the shape matters for feasibility.
5. **Weight error and the "optional rejection step" (§4.3).** Draws are exact from $\hat{\pi}$ (estimated weights), not π; the rejection step is not specified in §4.3 and omitted from the source's Algorithm 1; not implemented. The TV bound rests on the source's Monte Carlo assumptions, which are not verified here.
6. **Thin shells.** The source approximates the dimension by 10^5; the exact radial method is used instead.
7. **Infinite mixture / scale.** Only M shells are used; tested to $d = 5$ only; diffeomorphic transformations and MPI parallelism are not implemented.
8. **Outside this PR (sec. 2, not changed):** the level indices of eqs. (2)–(7) are inconsistent (eq. 3 uses Σ_0 where eq. 2 with $i = 1$ would use Σ_1; eq. 6 sums from $i = 0$ though eq. 5 defines ϵ_i for $i \geq 1$; by my derivation the gradient of eq. 6 is $-\partial\mathcal{F}/\partial v_i = J_i^\top \Sigma_i^{-1}\epsilon_i - \Sigma_{i+1}^{-1}\epsilon_{i+1}$, not eq. 7 as printed). Relevant for a future hierarchical predictive-coding layer; raised for discussion.

5 Validation

Tests (`tests/test_ellipsoidal.py`, 42 tests; full suite 69 passed in about 3 s): shell volumes (closed forms; shells add up to the ball), uniform sampling (membership, radial law, isotropy, thin shells, $d = 200$), input validation, exactness against closed forms (one shell; 1-D and correlated 2-D normal; 3-D with Monte Carlo bounds; bimodal mixture; mismatched partitions), the coalescence-time law, infeasible settings raise, reproducibility, and reproduction of the PR 1 conjugate posterior.

Experiment E4 (10 seeds, pooled draws, $\alpha = 0.001$, about 30 s): **45 of 45 checks pass** (39 positive, 6 negative controls that fail as required); exit status 0.

Part	Checked	Observed
A: one shell	KS vs exact CDF; $\mathbb{E}\|x\|$; $\mathbb{E}T$; violations	KS $p = 0.29$; $\mathbb{E}T = 1.179$ vs theory 1.181; 0 violations. Controls fail: uniform only; support $\{1, 2, \dots\}$
B: Gaussian, $d = 1, 2, 3, 5$	KS of Mahalanobis^2 vs χ^2_d; mean and covariance z-scores; violation rate	KS $p = 0.53/0.20/0.30/0.34$; all $\|z\| \leq 2.3$; 0 violations; $d = 2, 5$ also with rigorous bounds. Controls fail: uniform shell choice; tail truncated at $M = 4$
C: non-Gaussian, 1-D	bimodal mixture; Student-$t(5)$	KS $p > 0.001$ for both; mean coalescence 3.3 and 11.3. Control fails: central ellipsoid only
D1: PR 1 posterior	draws vs exact conjugate posterior	KS $p = 0.67$; $\|\bar{z}\| = 7 \times 10^{-4}$; $\|\mathrm{Var}\, z - 1\| = 3 \times 10^{-3}$
D2: SBC with the sampler	χ^2 on 10 rank bins; 50/80/95% coverage (3000 trials)	$p = 0.62$; coverage 0.498/0.784/0.944, all within 3 binomial SE. Control fails: wrong noise level

Two points about the experiment's own design. (a) One negative control was changed after the first run: truncating Student-$t(5)$ at 4 shells drops only $\sim 0.13\%$ of the mass, below what a KS

test can detect at 40,000 draws, so it could not fail; it was replaced by the central ellipsoid only ($\sim 5\%$). The $d = 2$ control ($M = 4$) is unchanged. (b) The SBC control must be wrong end to end (belief and likelihood), as in PR 1 E1. **Exit status:** verified by injecting the source's stated coalescence form into every sampler: 20 rows printed FAIL and the command exited 1.

6 What PR 2 claims, and what it does not

Claims (exactly these):

1. The implementation reproduces the paper's ellipsoidal decomposition sampler: with the coalescence time supported on $\{0, 1, \dots\}$ the perfect draw on a shell is exact, and the overall sampler matches closed-form references (Gaussian $d \leq 5$, a bimodal mixture, Student-$t(5)$) and the PR 1 conjugate posterior, including simulation-based calibration.
2. The documented discrepancy in item 1 is real in this implementation.

Does not claim: exactness of the draws from π itself (they are from $\hat{\pi}$); that the minorization always holds (estimated bounds); performance beyond $d = 5$ or for heavier tails than t_5; that the sampler is efficient for partitions whose shape differs strongly from the target's (p_i then collapses and the sampler raises); anything about the reflex loop, Thompson sampling, RGPs or optimisation.

Limitations. Across a shell the density varies by about $\exp(-(\beta_{\max} - \beta_{\min})\rho^2/2)$ however thin the shell is, so the partition's shape must be close to the target's (tested: mild mismatch works, strong anisotropic mismatch raises `SamplerInfeasibleError`); cost per draw is $O(1/\hat{p})$ residual steps (mean 1.2–2 for Gaussians, 11 for t_5); numpy only, no parallelism; single partition only (the multimodal extension is not implemented).

Remaining work (needs approval): Thompson sampling with exact draws; a belief class wrapping the sampler; recursive GP with the look-up table; derivative-aware optimisation (the paper only cites it).

7 Reproduce

```
pip install -r requirements.txt
pytest                                          # 69 passed
python experiments/e4_ellipsoidal_sampler.py    # exit 0, ends with PASS
```